Four Questions in Einstein Geometry: Conformal-Kähler Separation, the Einstein Weyl Channel, and the Complete Darboux Ricci Symbol

Mathematical continuation of the Qiushi 8 October handoff

8 October 2026

Abstract

We use the corrected statements in the two-session handoff of 8 October. The background-sensitive strict Burns saving is an input of that handoff. A scalar-energy argument extends the comparison barrier to all globally conformally Kähler metrics, without fixing their complex structure, and separates subthreshold minimizing sequences from that entire class. A concrete Horikawa blowup removes Einstein nonexistence from the list of unproved requirements for one possible minimizer example. An exact inversion of the recorded classical Weyl bound identifies the Weyl deficit forced on an Einstein competitor; it is a reformulation of the classical envelope, not an improved unrestricted lower bound. Finally, we calculate the full image, kernel, gauge coercivity, and a right inverse of the Darboux-compatible Ricci principal symbol. The kernel is exactly the Hamiltonian gauge direction. These statements provide concrete inputs for the four questions, but neither smooth global attainment nor the negative Goldberg conclusion follows from them.

1 The four questions and the source inputs

The present questions are those of [1, Section 1]: Einstein existence on $M = X\#k\overline{\mathbb{CP}}^2$ with X minimal of general type and $k \geq 1$; the unrestricted infimum of P; a smooth closed non-Bach-flat global minimizer of P; and compact negative Einstein almost-Kähler integrability in dimension four. The older O10/O11 labels are not substituted for these four specifications.

On a closed oriented four-manifold set

$$P(g) = \int_M |\operatorname{Ric}_g^\circ|^2 \, dV_g, \qquad S(g) = \int_M s_g^2 \, dV_g, \qquad W_\pm(g) = \int_M |W_g^\pm|^2 \, dV_g, \quad T_\pm = 2\chi \pm 3\tau. \tag{1.1}$$

Weyl norms are Hilbert–Schmidt norms on two-forms, while Ricci is a symmetric-tensor norm. The identities used throughout are

$$P = \frac{S}{12} + 4W_\pm - 8\pi^2 T_\pm, \qquad \int_M |\operatorname{Rm}_{\mathrm{op}}|^2 = 8\pi^2\chi + P. \tag{1.2}$$

Write $p_*(M) = \inf_{g \in \mathrm{Met}(M)} P(g)$; the normalized relative Einstein functional is not denoted by p_*.

For a minimal general-type model X let $A = c_1^2(X) > 0$. The classical scalar budget on its blowup is

$$S(g) \geq 32\pi^2 A \quad \text{for every smooth metric } g. \tag{1.3}$$

This follows, for example, from the classical scalar-curvature invariant of a general-type surface [3, Introduction]. Applications below may instead take (1.3) as their explicit input. The negative Yamabe property, when used, is stated separately.

The new overnight manuscript proves, in its stated geometric range,

$$p_*(X\#k\overline{\mathbb{CP}}^2) < 8\pi^2 k. \tag{1.4}$$

Its mechanism is the optimized Burns response

$$P(g_\varepsilon) = P(g) + 8\pi^2 - 8\pi^2\varepsilon\mathbf{R}_p^+(\zeta,\zeta) + 4\pi^2\Phi\varepsilon^2\log(1/\varepsilon) + O(\varepsilon^2), \tag{1.5}$$

with $\Phi = -|T|^2$ on an Einstein kernel direction. In the KE case the coefficient is $-s^2/32$ before multiplication by $4\pi^2$. We inherit (1.4) and its complete-support, background, matching, and resolution hypotheses from [1, Section 3], [2, A01.52 and corrected A02.59D]. None of the arguments below reproves the logarithmic construction.

2 An energy barrier for the entire conformally Kähler class

Let $\mathrm{CK}_+(M)$ be the set of smooth metrics globally conformal to a smooth Kähler metric inducing the chosen orientation of M. The complex structure and the positive conformal factor are both allowed to vary. Define $\mathrm{CK}_-(M)$ using the opposite orientation. This definition does not include merely locally conformally Kähler metrics or singular conformal factors.

Theorem 2.1. *Assume* (1.3). *If $g = u^2h$, with $u > 0$ smooth and h a smooth Kähler metric of orientation $\epsilon \in \{+,-\}$, then*

$$\boxed{P(g) - 8\pi^2(A - T_\epsilon) = \frac{S(g) - 32\pi^2 A}{12} + \frac{S(h) - 32\pi^2 A}{6} \geq 0.} \tag{2.1}$$

In particular, on $M = X\#k\overline{\mathbb{CP}}^2$ with its complex orientation,

$$\inf_{g\in \mathrm{CK}_+(M)} P(g) \geq 8\pi^2 k. \tag{2.2}$$

If additionally $\tau(M) \geq 0$, then

$$\inf_{g\in \mathrm{CK}_+(M)\cup \mathrm{CK}_-(M)} P(g) \geq 8\pi^2 k. \tag{2.3}$$

Proof. For a Kähler surface with its complex orientation, $|W_h^+|^2 = s_h^2/24$. This pointwise identity is classical [3, Introduction]. The integrated chiral Weyl energy is conformally invariant in dimension four. Consequently

$$W_\epsilon(g) = W_\epsilon(h) = S(h)/24.$$

Substitution in (1.2) gives

$$P(g) = S(g)/12 + S(h)/6 - 8\pi^2 T_\epsilon.$$

Subtract $8\pi^2(A - T_\epsilon)$ and use (1.3) for both metrics. This proves (2.1) without any assumption on how their complex structure relates to the original one on X. For a blowup, $T_+ = A - k$. If $\tau(M) \geq 0$, then $T_- \leq T_+$, so the opposite-orientation lower bound is at least as large. □

Corollary 2.2 (Separation of genuine minimizing sequences)**.** *Assume the input* (1.4) *and let g_{test} be a fixed smooth competitor satisfying*

$$P(g_{\mathrm{test}}) \leq 8\pi^2 k - \delta_0, \qquad \delta_0 > 0.$$

Every minimizing sequence $P(g_j) \to p_*(M)$ eventually satisfies

$$P(g_j) \leq 8\pi^2 k - \delta_0/2 \tag{2.4}$$

and therefore lies outside $\mathrm{CK}_+(M)$. If $\tau(M) \geq 0$, it eventually lies outside both conformally Kähler classes. The same exclusion applies to a smooth global minimizer whenever one exists.

Proof. The test metric gives $p_*(M) \le 8\pi^2 k - \delta_0$. Convergence of the energies gives the displayed bound for large j, while Theorem 2.1 excludes membership in the indicated classes. □

The separation is visible in a conformal invariant. For any smooth g with $P(g) \le 8\pi^2 k - \delta$, $\delta > 0$, equation (1.2) and the scalar budget imply

$$\boxed{W_+(g) \le \frac{4\pi^2 A}{3} - \frac{\delta}{4}, \qquad W_+(h) \ge \frac{4\pi^2 A}{3} \quad (h \in \mathrm{CK}_+(M)).} \tag{2.5}$$

Thus changing a positive conformal factor cannot turn the low-cost metric into a Kähler one. This is a separation in Weyl energy, not a claimed distance between metrics in a prescribed metric-space topology.

For a complex-hyperbolic X, $\tau(X) = A/9$, so the two-orientation statement applies when $k \le A/9$. The topology formulas are

$$T_+(M) = A - k, \qquad T_-(M) = A/3 + 5k, \qquad \tau(M) = A/9 - k.$$

The $+5k$ sign in T_- is essential.

What this contributes to Q2 and Q3. The handoff's strict upper bound and Theorem 2.1 give an energy separation from the entire indicated conformally Kähler locus, not merely from a fixed Kähler cone. The theorem is a short consequence of classical identities and the scalar budget; it is not represented as a new analytic coercivity theorem or as a proof of smooth attainment. No statement about arbitrary Hermitian or locally conformally Kähler metrics is inferred.

3 Exact Einstein candidates must overshoot the nominal Weyl threshold

In the source's negative Yamabe representative, put

$$\kappa = \frac{A}{2} - \frac{3W_+}{8\pi^2}, \qquad \upsilon = \frac{S}{32\pi^2} - A, \qquad p = \frac{P}{8\pi^2}. \tag{3.1}$$

These are exactly the Weyl-channel variables of [1, Section 7]; κ is not the O10 parameter defect. We use the two recorded relations

$$p = k - \frac{4}{3}\kappa + \frac{1}{3}\upsilon, \tag{3.2}$$

$$2\sqrt{A + \upsilon} + \sqrt{A - 2\kappa} \ge 3\sqrt{A}. \tag{3.3}$$

Dropping the nonnegative extra projection budget only weakens the second inequality. Its source is the classical combined scalar–Weyl estimate [3], not a new lower bound.

Proposition 3.1 (Inverting the classical feasible region)**.** *Let $\ell = k - p$ and suppose $0 < \ell < A/3$. Every competitor satisfying (3.2)–(3.3) obeys*

$$\boxed{\kappa \ge r_-(\ell) := \frac{2A + 6\ell - 2\sqrt{A^2 - 3A\ell}}{9} = \ell + \frac{\ell^2}{A(1 + \sqrt{1 - 3\ell/A})^2}.} \tag{3.4}$$

Consequently, an Einstein metric in the range $0 < k < A/3$ must have

$$\boxed{\kappa \ge k + \frac{k^2}{A(1 + \sqrt{1 - 3k/A})^2} > k.} \tag{3.5}$$

Proof. If $\kappa \le 0$, then $\upsilon \ge 0$ and (3.2) give $p \ge k$, contrary to $\ell > 0$. For $0 \le \kappa \le A/2$, solving (3.3) for υ gives the source bound

$$p \ge L_A(k,\kappa) = k - \kappa + \frac{1}{2}\{A - \kappa - \sqrt{A^2 - 2A\kappa}\}.$$

This function is strictly decreasing up to $\kappa = 4A/9$ and then increasing. Its smaller intersection with $p = k - \ell$ solves

$$\sqrt{A^2 - 2A\kappa} = A - 3\kappa + 2\ell,$$

whose smaller root is the first expression for r_-. For $0 < \ell < A/3$, this root lies in $(0, 4A/9)$ and the unsquared right side is positive there. Thus $\kappa < r_-(\ell)$ is impossible. Any $\kappa > 4A/9$ already satisfies the desired lower bound. Writing $t = \sqrt{1 - 3\ell/A}$ gives

$$r_-(\ell) - \ell = \frac{A}{9}(1-t)^2 = \frac{\ell^2}{A(1+t)^2},$$

which proves the second expression. For Einstein $P = 0$, so $\ell = k$. □

This calculation fixes a precise target for an *Einstein-specific* upper estimate: it would be enough to prove $\kappa < r_-(k)$ on those exact Einstein candidates. Such an upper estimate has not been obtained here. The stronger requirement $\kappa \le k$ for *all* metrics would imply it but is not necessary for Q1. This distinction avoids making a full Q2 lower-bound theorem a prerequisite for every approach to Q1.

The restriction is still only classical feasibility. For example, for $0 < k/A \le 1/4$ the formal data

$$P = 0, \qquad \kappa = 2k, \qquad \upsilon = 5k \tag{3.6}$$

satisfy both (3.2) and (3.3). To check the latter, write $h = k/A$; it is $2\sqrt{1+5h} + \sqrt{1-4h} \ge 3$, which, after one legitimate squaring, reduces to $\sqrt{1-4h} \ge 1 - 4h$. They are not a metric or an actual-field counterexample. They show exactly why inversion of the old budget cannot alone answer Q1.

4 The attainment requirement in Q3

Put $\mathcal{W}(g) = W_+(g) + W_-(g)$ and let $\mathfrak{B}$ denote its L^2 gradient, with respect to variations of the covariant metric. It is a fixed nonzero multiple of the Bach tensor. This definition avoids mixing normalizations of the Bach tensor across references.

From (1.2),

$$P = 2\mathcal{W} + S/12 - 16\pi^2\chi. \tag{4.1}$$

For a conformal variation $g_t = e^{2tf}g$ in dimension four,

$$\left.\frac{d}{dt}\right|_{t=0} P(g_t) = -\int_M f\,\Delta s\,dV, \tag{4.2}$$

where $\Delta = \nabla^i\nabla_i$. Thus a smooth closed critical metric has constant scalar curvature. At such a metric, $\nabla S = -2s\,\mathrm{Ric}^\circ$; hence criticality of (4.1) gives

$$\boxed{\mathfrak{B} = \frac{s}{12}\,\mathrm{Ric}^\circ\,.} \tag{4.3}$$

These identities follow as well from the full quadratic-curvature variation formulas in [5, Section 2].

On a manifold with negative Yamabe invariant a constant-scalar metric has $s < 0$. Equation (4.3) then yields

$$\text{smooth } P\text{-critical and Bach-flat} \quad\Longleftrightarrow\quad \text{Einstein.} \tag{4.4}$$

Therefore a valid route to a Q3 example on one chosen M is

$$\text{smooth attainment on } M + \text{Einstein nonexistence on } M + \mathcal{Y}(M) < 0.$$

Theorem 2.1 excludes conformally Kähler competitors below the standard barrier, but it proves neither of the first two requirements. Anderson's selected-minimizer framework [6] does not turn orbifold or partly collapsed convergence into smooth attainment on the original differentiable manifold without the further connections specified in [1, Section 4].

4.1 A concrete candidate that does not depend on unresolved small-blowup nonexistence

For an existence question such as Q3, the background need not be a large ball quotient whose one-point blowup lies in the unresolved Q1 range. The following classical surface gives a concrete alternative. Let

$$X = \{w^2 + z^5 + x^{10} + y^{10} = 0\} \subset \mathbb{P}(1,1,2,5), \qquad M = X\#\overline{\mathbb{CP}}^2. \tag{4.5}$$

The coordinates have weights $(1,1,2,5)$. The affine cone is smooth away from the origin, since the four partial derivatives vanish simultaneously only there. The hypersurface avoids the two singular points of the weighted projective space, and hence is a smooth surface. Weighted adjunction gives $K_X = \mathcal{O}_X(1)$, an ample line bundle, with

$$K_X^2 = \frac{10}{1\cdot 1\cdot 2\cdot 5} = 1, \qquad p_g(X) = 2, \qquad q(X) = 0.$$

These are the classical $(1,2)$ Horikawa surfaces; their weighted hypersurface description and invariants are recalled in [4, Introduction]. In particular X has a negative KE metric, and its unperturbed monopole supply is in the $b^+ > 1$ range. Noether's formula and the signature formula give

$$\chi(X) = 35, \quad \tau(X) = -23, \qquad \chi(M) = 36, \quad \tau(M) = -24, \quad T_+(M) = 0. \tag{4.6}$$

The all-metric scalar and combined scalar–Weyl budgets for the blowup use $A = 1$. The classical parameter inequality at $\beta = 8$ is

$$D_8 = \frac{27}{2}P + 108\pi^2 T_+ - 72\pi^2 p_{\mathrm{SW}} \geq 0, \qquad p_{\mathrm{SW}} \geq 1.$$

Thus the source's general-KE strict upper bound gives the explicit interval

$$\boxed{\frac{16\pi^2}{3} \leq p_*(M) < 8\pi^2.} \tag{4.7}$$

The lower bound is classical. The strict upper bound is the overnight construction (1.4), applied to this fixed smooth negative KE background. This combination does not determine $p_*(M)$.

Since $p_*(M) > 0$, this M carries no Einstein metric. It also has negative Yamabe invariant by the standard general-type scalar theory. Consequently

$$\boxed{\text{If } P \text{ attains its infimum smoothly on this } M\text{, then the minimizer is non-Bach-flat.}} \tag{4.8}$$

Theorem 2.1 additionally excludes globally conformally Kähler minimizers inducing the displayed orientation. No opposite-orientation conclusion is asserted here, since $\tau(M) < 0$.

This choice removes the unresolved Q1 Einstein alternative from this one Q3 attempt. What remains is a genuine attainment problem. The strict upper bound excludes a separate standard Burns loss of energy $8\pi^2$ under the stated additive-energy hypotheses, but does not exclude cheaper ALE channels, Ricci-flat bubbles, or partial collapse. Therefore (4.8) is not an example of an attained minimum. Its value is to isolate attainment on a concrete smooth manifold using already available nonexistence and upper-bound information.

5 The complete Darboux Ricci principal symbol

This section supplies a concrete replacement for an ill-conditioned local jet block. It works at a single point and nonzero real cotangent vector; it does not assert formal integrability to all orders.

Let (V, g, J) be a Hermitian real four-space. Identify a symmetric tensor H with its g-symmetric endomorphism and set

$$\mathcal{E}_J = \{H : HJ + JH = 0\}, \qquad \dim_{\mathbb{R}} \mathcal{E}_J = 6.$$

For a real covector ξ, identified with its dual vector, define the Ricci symbol in the source's convention by

$$S_\xi(H) = \frac{1}{2}\{\xi \otimes H\xi + H\xi \otimes \xi - |\xi|^2 H\}. \tag{5.1}$$

A simultaneous change of sign does not affect the statements about kernel, image, or norms.

Theorem 5.1. *For every $\xi \neq 0$,*

$$\ker S_\xi = \mathbb{R}(\xi \otimes J\xi + J\xi \otimes \xi), \tag{5.2}$$

$$\operatorname{im} S_\xi = \left\{K \in \operatorname{Sym}^2 V^* : K\xi - \tfrac{1}{2}(\operatorname{tr} K)\xi = 0, \quad K(J\xi, J\xi) - \tfrac{1}{2}|\xi|^2 \operatorname{tr} K = 0\right\}. \tag{5.3}$$

In particular the image has dimension five: the four contracted-Bianchi conditions and the additional scalar Darboux condition are the complete linear conditions at this fixed real covector. Moreover

$$\boxed{|S_\xi(H)|^2 + \frac{1}{4}|H(\xi, J\xi)|^2 \geq \frac{1}{8}|\xi|^4|H|^2.} \tag{5.4}$$

The constant $1/8$ is sharp.

Proof. By unitary covariance and homogeneity, it is enough to take $\xi = e_1$, $Je_1 = e_2$, $Je_3 = e_4$. Every $H \in \mathcal{E}_J$ has the form

$$H = \begin{pmatrix} a & b & c & d \\ b & -a & d & -c \\ c & d & e & f \\ d & -c & f & -e \end{pmatrix}. \tag{5.5}$$

Direct substitution gives

$$S_{e_1}(H) = \frac{1}{2}\begin{pmatrix} a & 0 & 0 & 0 \\ 0 & a & -d & c \\ 0 & -d & -e & -f \\ 0 & c & -f & e \end{pmatrix}. \tag{5.6}$$

The only absent variable is b, which proves (5.2) and the rank-five statement. Conversely, the equations on the right of (5.3) force $K_{12} = K_{13} = K_{14} = 0$, $K_{11} = K_{22}$, $K_{44} = -K_{33}$. These leave precisely the five entries in (5.6), so there is no further image condition at this covector. Finally,

$$\begin{aligned} |S_{e_1}(H)|^2 &= \tfrac{1}{2}(a^2 + c^2 + d^2 + e^2 + f^2), \\ |H|^2 &= 2(a^2 + b^2 + e^2 + f^2) + 4(c^2 + d^2), \\ H(e_1, Je_1) &= b. \end{aligned}$$

Consequently the difference between the left side and the right side of (5.4) is exactly $\frac{1}{4}(a^2 + e^2 + f^2) \geq 0$. Equality holds, for example, when only c is nonzero. This proves sharpness and the theorem. □

5.1 An explicit image projection and right inverse

For a general symmetric matrix K at $\xi = e_1$, its orthogonal projection onto (5.3) is

$$\mathbb{P}K = \begin{pmatrix} t & 0 & 0 & 0 \\ 0 & t & K_{23} & K_{24} \\ 0 & K_{23} & u & K_{34} \\ 0 & K_{24} & K_{34} & -u \end{pmatrix}, \qquad t = \frac{K_{11}+K_{22}}{2}, \quad u = \frac{K_{33}-K_{44}}{2}. \tag{5.7}$$

The gauge-fixed right inverse $\mathbb{R}$ on the image is obtained in (5.5) by

$$a = 2K_{11}, \quad b = 0, \quad c = 2K_{24}, \quad d = -2K_{23}, \quad e = -2K_{33}, \quad f = -2K_{34}. \tag{5.8}$$

It satisfies $S_{e_1}\mathbb{R}K = K$ on the image and $|\mathbb{R}K| \le \sqrt{8}|K|$. For unit ξ, the squared singular values of S_ξ are

$$0, \quad \tfrac{1}{8}, \quad \tfrac{1}{8}, \quad \tfrac{1}{4}, \quad \tfrac{1}{4}, \quad \tfrac{1}{4}. \tag{5.9}$$

These formulas permit explicit treatment of the image and gauge rather than repeatedly inferring them from a nearly singular numerical Jacobian.

5.2 Hamiltonian meaning and the nonlinear metric parametrization

For a fixed Darboux symplectic form, Hamiltonian vector fields have symbol $X_f = J\xi f$; the metric variation $\mathcal{L}_{X_f} g$ has principal part $\xi \otimes J\xi + J\xi \otimes \xi$, up to the Fourier sign. Thus the kernel in (5.2) is genuine gauge freedom, not evidence of a strictly almost-Kähler Einstein deformation. A scalar gauge with principal symbol $H(\xi, J\xi)$ removes it; its action on the Hamiltonian potential has a nonzero multiple of $|\xi|^4$ as symbol.

There is also a direct way to remove the algebraic metric constraints. In a fixed Darboux frame choose $J_0^2 = -I$ and use

$$G = \exp H, \qquad H^T = H, \qquad HJ_0 + J_0 H = 0. \tag{5.10}$$

Then G is positive definite, $\det G = 1$, and

$$GJ_0G = J_0. \tag{5.11}$$

The compatible metric constraint and closure of the fixed symplectic form are exact; they need not be reintroduced as soft numerical residuals. This supplies six unconstrained coefficient functions before Hamiltonian gauge fixing. It is a local parametrization, not an imposed global trivialization of the tangent bundle.

5.3 What the symbol calculation does not imply

The extra Darboux compatibility relation and the correction of the degree-15 count are already in [2, A01.55]. Theorem 5.1 adds the full fixed-covector kernel/image calculation, explicit right inverse and norm control. It does not show that a nonlinear lower-jet source lies in the prolonged image, that the system is involutive, or that a formal solution converges. Curvature sign is absent from this principal symbol; an obstruction distinguishing negative Einstein data from KE and Ricci-flat controls must enter lower-order resolved compatibility or global geometry. The classical formal-integrability perspective is discussed in [7, Section 4].

5.4 A reproducible highest-order block calculation

The accompanying script constructs the integer matrix of the homogeneous highest-order operator $2D\,\mathrm{Ric}_\delta$ from degree-N compatible tensor polynomials to degree-$(N-2)$ symmetric tensor polynomials. A run in this turn gave the following ranks over both primes 65521 and 65519:

N	Rows	Columns	Rank modulo each prime	Kernel minus Hamiltonian count
4	100	210	83	43
5	200	336	156	60
6	350	504	260	79
7	560	720	400	100

These agree with the corrected count $N^2 + 8N - 5$ already recorded in [2, A01.55]. The degree-seven integer matrix occupies 3,225,600 bytes in this implementation. Agreement modulo two primes is a check of these finite matrices; it is not alone a proof of the rational rank in every degree or of nonlinear prolongation compatibility.

The computation is deliberately different from solving the complete nonlinear finite-jet equations: the known linear image, Hamiltonian gauge, and compatibility conditions are separated first. At each degree a new nonlinear forcing term must still lie in the prolonged image. Testing that membership, and calculating any resulting lower-order condition, cannot be replaced by the fixed-covector inverse above.

6 Consequences for the four research tasks

The following dependencies keep the four questions distinct.

1. A positive unrestricted lower bound for P on one M excludes its Einstein metrics. Q1 can also be approached directly at $P = 0$, without first computing $p_*(M)$. Proposition 3.1 gives the exact classical Weyl threshold that such an argument must improve.
2. The strict upper bound in the handoff, together with Theorem 2.1, places low-energy competition outside an entire conformally Kähler class. A matching lower bound remains necessary to determine p_*; a conformal modification cannot remove this competition.
3. Q3 requires smooth attainment and exclusion of the Einstein alternative on the same smooth manifold. For the concrete manifold (4.5), the Einstein alternative is already excluded and (4.7) gives an explicit energy window. Smooth attainment remains unproved; the Bach identity is an implication, not an existence theorem.
4. Q4 still needs a new resolved global or lower-order compatibility constraint, or a complete closed counterexample. The complete symbol calculation changes the numerical formulation and prevents counting gauge and known identities as mathematical obstructions.

No result here declares Q1, Q2, Q3, or Q4 fully solved. The additions are the complete conformally Kähler separation argument, the explicit inversion of the source's classical envelope, and the gauge-resolved Ricci-symbol calculation. The general-background negative logarithmic construction remains the system's result, and the scalar, curvature-variation and symbol background remains classical. These provenance distinctions are part of the mathematical statement, not separate research objectives.

References

[1] Qiushi Sessions S1003A01 and S1003A02, *Einstein Gaps on Four-Manifolds: Four Public Problems*, 8 October 2026, supplied file `01_Research_Synthesis(1).tex`. The sections on the four tasks, the insertion theorem, minimizers, Weyl coordinates, and Goldberg compatibility specify the four tasks, the corrected Burns theorem, attainment conditions, Weyl-channel bounds, and Goldberg corrections.

[2] Same sessions, *Mathematical Source Companion*, supplied file `02_Mathematical_Source_Companion(1).tex`. Source keys A01.52, A01.55, A02.58, A02.59D, A02.62–64.

[3] C. LeBrun, *Ricci curvature, minimal volumes, and Seiberg–Witten theory*, Invent. Math. 145 (2001), 279–316. https://arxiv.org/abs/math/0003068.

[4] S. Coughlan and R. Pignatelli, *Simple fibrations in* $(1,2)$*-surfaces*, Forum of Mathematics, Sigma (2023). https://doi.org/10.1017/fms.2023.41. The introduction recalls the classical Horikawa surface description; no new surface-existence theorem is claimed here.

[5] M. J. Gursky and J. A. Viaclovsky, *Rigidity and stability of Einstein metrics for quadratic curvature functionals*, arXiv:1105.4648v2 (2013). https://arxiv.org/abs/1105.4648.

[6] M. T. Anderson, *Canonical metrics on 3-manifolds and 4-manifolds*, Asian J. Math. 10 (2006), 127–163. https://arxiv.org/abs/math/0511034.

[7] V. Apostolov and T. Drăghici, *The curvature and the integrability of almost-Kähler manifolds: a survey*, 2003. https://arxiv.org/abs/math/0302152.

[8] V. Apostolov, J. Armstrong and T. Drăghici, *Local models and integrability of certain almost Kähler 4-manifolds*, Math. Ann. 323 (2002), 633–666. https://arxiv.org/abs/math/0109098.